

Assignment 4

Date: 25/8/2026

1. Create a smart-home application through Python that manages devices such as lights, fans, and air conditioners.

Create a class 'SmartDevice' with information about the device name, room, power consumption, and current status. [Constructor, Instance Attributes]

The system should:

- Prevent invalid power-consumption values.
- Allow the device to be switched ON or OFF.
- Calculate the estimated power usage based on the number of operating hours.
- Display a readable description whenever the object is printed. `__str__()`
- Provide a detailed representation suitable for inspecting objects inside a list. `__repr__()`

The application should also maintain the total number of devices registered in the smart-home system. [Class Attribute]

Create at least four devices and demonstrate the effect of modifying their individual status and the shared device count.

2. Python Programme for software application temporarily creates objects that represent system resources, such as database connections or application sessions. Create a class Resource that stores: Resource name, Resource type, Resource ID.

When an object is created, display a message indicating that the resource has become active.

When an object is destroyed, display a message indicating that the resource has been released. `__del__()`

The class should also:

- Reject an empty resource ID. [assert]
- Provide a meaningful display when printed. `__str__()`
- Provide a detailed representation for debugging and collection display. `__repr__()`

Create several resources, explicitly remove selected objects using del, and observe the resulting lifecycle messages.

3. WAP for -

A university system may create different Python objects representing the same student. The system needs a meaningful way to determine whether two student objects refer to the same student.

Create a class `StudentProfile` containing: Student ID, Name, Program, Semester.

Two student objects should be considered equal when they have the same student ID, even if their other information is different. **`__eq__()` – learn the dunder method and use**

Also provide:

- A readable output when an individual object is printed. `__str__()`
- A detailed representation when objects are inspected inside a collection. `__repr__()`

Create at least four student objects and compare different pairs using `==`.

WITHOUT REFERRING TO METHODS THAT WILL HAVE TO BE USED.

4. Develop E-Commerce Product Catalogue by python prog. Lang. for an e-commerce platform maintains a catalogue of products sold by different sellers.

Create a class `Product` containing: Product ID, Product name, Category, Price, Stock quantity

The application should:

- Reject invalid prices and stock quantities.
- Allow stock to be increased or decreased.
- Calculate the current inventory value of a product.
- Maintain a common platform name for all products.
- Keep track of the number of product objects registered through a class-level mechanism.
- Provide meaningful output when individual products are printed.
- Provide detailed representations when a collection of products is displayed.

Task: Create at least five products, modify stock for multiple products, display their inventory values, and demonstrate the difference between class-level and object-level information.

5. WAP where a university is developing a system to manage different types of digital resources used by students. Each resource has a name, owner, size, and unique resource ID.

Design a class `DigitalResource` that represents such a resource.

The system should satisfy the following requirements:

- Every resource must be created with valid identifying information.

- The resource size must be greater than zero.
- A resource can be updated after creation.
- The university name should be common to all resources.
- Printing a resource should provide a clear, user-friendly description.
- Printing a collection of resources should provide enough information to identify each object.
- When a resource is explicitly removed, the program should report that the resource has been released.
- The program should demonstrate the comparison of two resources based on their unique IDs.

Task:

1. Create the class according to the requirements.
2. Create at least five resource objects.
3. Modify selected resources.
4. Compare resources with the same and different IDs.
5. Display individual resources.
6. Store the resources in a list and display the collection.
7. Change the common university name and observe the result.
8. Test invalid resource information.
9. Explicitly delete selected resources and observe the object-lifecycle message.